

NWIS

WELL COMPLETION REPORT

Doc ID
W002-WCR-01

Well: Alpha-02 | Well ID: W002 | Field: Field_Alpha

PRIMARY ISSUE CATEGORY: STUCK PIPE

1. GENERAL WELL INFORMATION

Well Name	Alpha-02	Well ID	W002
Field	Field_Alpha	Block	Block_1
Spud Date	10-09-2024	Rig Release Date	12-10-2024
Total Depth (TD)	3320 m	Rig Name	Rig Sagar-3
Well Type	Directional Development	Trajectory	S-shaped (build 0-28 deg by 2600 m, hold to TD)
Max Inclination	28 deg	Directional Interval	2350-2600 m (build section)

2. FORMATION & PORE PRESSURE INFORMATION

Formation_Y (reactive, over-consolidated shale) was encountered between 3080 m and 3260 m. This shale is known to swell on contact with water-based fluids and is associated with hole-cleaning difficulty and pack-off tendencies in the 28-deg tangent section across Field_Alpha directional wells.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2700-3080	Formation_X (lower)	Sand/shale interbedded	Normal drilling trend
3080-3180	Formation_Y (upper)	Shale, reactive, laminated	Swelling tendency; hole cleaning risk
3180-3260	Formation_Y (lower)	Shale, reactive, fissile	Primary stuck-pipe interval
3260-3320	Formation_Z	Shale, dark grey, firm	Normal drilling trend

4. HISTORICAL DRILLING / OPERATIONS EVENTS

Event 1: Stuck Pipe (Pack-Off / Differential Sticking) at approximately 3160 m

Observation: Overpull increased progressively from 15 klb to 55 klb over three connections; torque trended up from 9 kft-lb to 16 kft-lb. String became immobile (no rotation, no reciprocation) while attempting to ream back to bottom after a connection in the 28-deg tangent section.

Engineering Action: Circulation was maintained at maximum safe rate, a 40 bbl hi-vis sweep was pumped to lift cuttings bed, and a free-point indicator (FPI) survey was run confirming free pipe to 3105 m. A diesel-based pipe-freeing pill (20 bbl) was spotted across the pack-off interval and the string was worked with jarring at 60-80 klb overpull cycles.

Outcome: String was freed after approximately 9 hours of remedial work. Drilling resumed with an increased hi-vis sweep frequency (every 150 m) and reduced ROP through the remainder of the reactive shale interval.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
Conductor	20	58	Surface

Surface	13-3/8	640	Surface
Intermediate	9-5/8	2600	1300
Production	7	3320	2400

6. NON-PRODUCTIVE TIME (NPT) SUMMARY

Issue	Depth (m)	NPT (hrs)	Category
Stuck pipe (pack-off) at 3160 m	3160	9.0	Wellbore Instability / Stuck Pipe

7. ROOT CAUSE ANALYSIS

Root cause analysis attributes the sticking event to inadequate hole cleaning in the 28-deg tangent section combined with reactive shale swelling that reduced effective annular clearance. Cuttings accumulated on the low side of the hole (cuttings bed) during a period of reduced flow rate at connection, leading to a mechanical pack-off around the BHA. Torque and overpull trends in the 24 hours prior to the event show a clear escalating pattern that, in hindsight, should have triggered a preventive wiper trip.

8. LESSONS LEARNED

- Escalating torque (>1.5x baseline) combined with rising connection overpull is a leading indicator of pack-off in this shale, typically 8-12 hours ahead of a stuck-pipe event.
- Hi-vis sweeps every 150 m (rather than every 300 m) in the 25-30 deg tangent section materially reduce cuttings-bed buildup.
- A preventive short wiper trip when torque exceeds 1.4x baseline in Formation_Y is cheaper than a stuck-pipe recovery.
- Diesel-based freeing pills were effective within the 3080-3260 m reactive shale interval; keep pre-mixed volume on standby for any well drilling this formation.

9. RECOMMENDATION FOR OFFSET WELLS

For offset wells drilling Formation_Y through the tangent section, implement real-time torque/overpull trend alarms (1.4x and 1.7x baseline thresholds), increase hi-vis sweep frequency to every 150 m between 3080-3260 m, and schedule a preventive wiper trip if torque trend persists for more than 3 connections.

This Well Completion Report is a synthetically generated historical record produced for the NWIS prototype (W002). Not associated with an actual drilling operation.